

Customer and Sales Data Analysis Report

Executive Summary

This project analyzes customer and sales data to identify customer behavior, sales trends, regional performance, and product performance. The analysis was performed using Python, Pandas, and Matplotlib.

Objectives

- Clean and prepare customer and sales datasets.
- Analyze customer churn and contract preferences.
- Identify top-performing products.
- Analyze regional sales distribution.
- Create visual dashboards for business insights.

Data Cleaning

The following preprocessing steps were performed:

- Removed unnecessary spaces from column names.
- Filled missing numerical values using median values.
- Filled missing categorical values with "Unknown".
- Converted sales dates into datetime format.
- Removed invalid records.

Customer Analysis

Customer Lifetime Value

Customer Lifetime Value (CLV) was analyzed using the TotalCharges column to estimate the revenue generated by each customer.

Churn Analysis

The churn rate was calculated to determine the percentage of customers who discontinued the service.

Contract Analysis

Customer contracts were categorized into different contract types to understand customer preferences.

Payment Method Analysis

Payment methods were analyzed to identify the most preferred transaction methods.

Sales Analysis

Monthly Sales Trend

Monthly sales were aggregated to identify growth patterns and seasonal variations.

Best-Selling Products

Products were ranked according to quantity sold and total revenue generated.

Regional Sales Performance

Sales were analyzed across different regions to identify high-performing markets.

Advanced Analysis

A pivot table was created to summarize sales performance by region and product category.

Dashboard Visualizations

The following visualizations were generated:

1. Monthly Sales Trend
2. Top Products by Revenue
3. Regional Sales Distribution
4. Contract Distribution
5. Churn Distribution

Key Findings

- Certain products generated significantly higher revenue than others.
- Regional sales varied across different locations.
- Customer churn affected overall customer retention.
- Contract preferences indicated customer commitment levels.

Business Recommendations

- Focus marketing efforts on top-performing products.
- Develop retention strategies to reduce churn.
- Strengthen operations in high-performing regions.
- Offer incentives for long-term contracts.
- Optimize inventory based on product demand.

Conclusion

The analysis provides valuable insights into customer behavior and sales performance. The generated visualizations and reports can assist management in making data-driven business decisions.